Roadmap: from chip to an embodied, closed-loop fly

What NeuroMPS builds next, after the final round. The plan below is a commitment, not a result: nothing in it has been run yet except where a number cites `outputs/metrics.json`. Everything it will produce is simulated, for research decision support; none of it is a claim about any compound's effect in any animal or person.

Slide message: *Today we read a simulated fly brain as a chip and as an animal, through the neurons that drive behaviour. Next we give it a body and close the loop, so that the question "does the chip flag a drug before behaviour fails?" can be asked of movement itself, with sensory feedback in place. Each stage has a registered go/no-go test.*

1. Why close the loop

The chip-to-animal study (report Act VII) reads behaviour as the firing of named neurons, down to the motor neurons of the male nerve cord. Three things it cannot see:

- **Movement.** A motor neuron firing faster is not a leg moving differently. Muscles, joints and contact with the ground turn firing into behaviour, and they can mask or amplify a change.
- **Feedback.** In a real fly, movement is sensed and fed back: leg proprioceptors report joint angles and loads to the nerve cord and brain. A drug that disturbs a circuit inside a loop can have effects an open-loop model misses, in either direction.
- **The environment.** Feeding is a sequence (find, taste, extend the proboscis), not a firing rate; grooming is a sequence of leg sweeps.

Our current results are the baseline the embodied model has to be compared against. On the simulated fly brain, the chip's firing rate ranks drugs by how much they disrupt behaviour (Spearman 0.863), but it raises the alarm late. At the motor neurons of the male nerve cord, the chip is later than motor output in 4 drug families, level in 1, and never earlier. And electrodes on the brain's output tract caught GABA block in time where a planar array did not (0.361 against 0.044 log2). The embodied model asks whether these conclusions survive movement and feedback.

2. What already exists

Layer	Status	Where
Whole-brain model (FlyWire), drug knobs by transmitter class	built, validated against the published sugar→MN9 result	`src/neuromps/fly/model.py`; report §23
Brain and nerve cord (MaleCNS, 165,122 neurons)	built; replicates the behaviour results	`fly/malecns.py`; §25
Virtual MEA scored by the chip's own extractor	built	`fly/electrodes.py`, `features/mea.py`
Motor-neuron readouts (proboscis, front legs, all motor neurons)	built, registered study done	`fly/motor.py`; §25
Neuron-to-muscle map	in the data: 260 of 381 leg motor neurons are typed by the muscle they drive	`fly/roadmap_inventory.py`
Feedback neurons	in the data: 1,030 proprioceptive sensory neurons in the nerve cord, among them 403 chordotonal-organ and 175 campaniform-sensilla neurons	`fly/roadmap_inventory.py`
Body and physics	available, not yet used: NeuroMechFly v2 (FlyGym) and flybody, both on MuJoCo, all Apache-2.0	external

3. The loop we will close

1. **Brain and nerve cord** (MaleCNS, leaky integrate-and-fire, the drug knobs as now).
2. **Motor neurons to muscles.** Each leg motor neuron typed by its muscle drives that muscle's activation; the untyped ones are mapped by their connectivity to typed partners, or left out and reported.
3. **Muscles to joints.** Muscle activations become joint torques in the body model (NeuroMechFly v2's leg joints).
4. **Body and world.** MuJoCo integrates the body against the ground, with leg adhesion, and against a sugar spot for feeding.
5. **Joints back to neurons.** Joint angles and contact forces drive the matching proprioceptive sensory neurons (chordotonal organs for angles, campaniform sensilla for load), which re-enter the network.

The chip side does not change: the same virtual array, scored by the same extractor, read while the fly behaves.

4. Stages, each with a registered go/no-go test

Stage	When	Deliverable	Go/no-go test (registered before it runs)
1. Open-loop legs	weeks 1–3	front-leg motor neurons drive NeuroMechFly's front legs	antennal drive produces front-leg sweeps toward the head that untreated, undriven runs do not
2. Feeding and walking	weeks 3–6	proboscis extension from the proboscis pool; walking from descending commands	sugar drives proboscis extension and bitter suppresses it; a walking command produces an alternating gait
3. Close the loop	weeks 6–10	proprioceptive feedback into the nerve cord	feedback measurably changes the movement (it stabilises or reshapes the rhythm) against the open-loop run
4. Drugs in the loop	weeks 10–14	the registered chip-to-animal design rerun with movement as the readout	registered predictions, among them: does the chip still flag drugs after movement fails; does the output-tract electrode still lead
5. Check against real flies	weeks 14–18	comparison with published fly pharmacology (feeding and grooming under GABA-A and GluCl drugs)	the direction of each published behavioural change is reproduced, or the mismatch is reported
6. Release	weeks 18–20	open code, models and a paper	one command reproduces every number, as now

Each stage stops if its test fails, and the failure is reported, as every failed prediction in this project has been.

5. What it would change for organ-on-a-chip

The embodied model lets three chip-design questions be asked of behaviour itself rather than of neurons:

- **Sensitivity.** How small a firing-rate change must a chip resolve to flag each mechanism before movement fails? Today's answer comes from interneurons and motor neurons (report §24).
- **Electrode placement.** Does recording the connection between compartments (the output tract, a chip's microchannels) still lead a planar array when behaviour is movement under feedback?
- **Screen thresholds.** How does a regulatory screen's hit cutoff compare with the change at which movement, not firing, fails?

6. Risks, and how each is handled

Risk	Handling
Part of the neuron-to-muscle map is missing (motor neurons not typed by muscle)	map the rest by connectivity to typed partners; report coverage; test robustness by leaving the uncertain ones out
A leaky integrate-and-fire nerve cord may not generate rhythms (walking needs them)	stage 2's go/no-go test catches it; then add the intrinsic currents rhythm generation needs, as a stated change
The male nerve cord is over-excitable under the published weight (any drive recruits many motor neurons)	calibrate one weight against the untreated motor recruitment before stage 4, registered
Real fly pharmacology is sparse and in different units	stage 5 compares directions, not magnitudes, and says which comparisons were impossible
Compute	FlyGym 2 simulates on GPU; the GB10 runs the network and the body in parallel

7. Status, and what stage 1 changed

Stage 1 ran and failed its go/no-go test, as did its registered follow-up. The male-CNS network was wired to the FlyMimic leg (MuJoCo 3.16, the leg's 15 Hill-type muscles driven by 49 of the 68 left front-leg motor neurons, by name). Antennal drive moved the claw toward the head in every trial, but sugar drive moved it closer still (median closest approach 0.9386 against 1.1067 mm): the movement was not specific. Stage 1b rescaled the male-CNS weight by the ratio of synaptic input per neuron between the connectomes; at that weight sugar no longer drove the feeding motor neuron at all (0.0 Hz), so its leg results are uninterpretable, as registered.

What went wrong. The same sugar drive puts 251 neurons above 10 Hz in the published FlyWire model and 14,263 in the male CNS at the same weight, most of them central-brain neurons (11,082 of 32,160); at the rescaled weight still 7,586. The published model's parameters do not carry over to the second connectome's brain: there, activity avalanches, and lowering one global weight silences weak pathways before it stops the avalanche. This is itself a result for the paper, and it changes the plan.

The revised plan. Use each connectome where its model is validated:

- **Stage 1c (next):** drive the male nerve cord from its descending neurons alone, without the male brain. Activating the descending neurons the literature ties to antennal grooming should move the front leg toward the head; descending neurons for other behaviours should not. This tests the nerve cord, the muscle map and the leg without the brain's avalanche.
- **Then:** join the FlyWire brain, where the published model is validated, to the male nerve cord through descending neurons matched by cell type across the two connectomes.

Both steps are registered before they run, like every stage.

Stage 1c ran: three of five registered predictions pass. With the male brain removed and named descending neurons driven, the nerve cord is silent without drive, and the descending neurons the literature ties to antennal grooming (aDN1, aDN2) drive the left front-leg motor pool at 15.846 Hz, against 2.72 for the middle and 1.697 for the hind leg; no other descending neuron recruits the front leg as strongly (forward walking 3.601 Hz). The kinematic prediction failed: grooming drive brings the claw to 1.1345 mm of the antenna, but forward-walking drive reaches 1.154 mm at the low points of a rhythmic swing, so the closest approach does not separate the two. The figure shows the difference the readout missed: under grooming drive the claw goes to the head and stays there; under forward-walking drive it oscillates. That is the readout to register next (stage 1d, fresh seeds), not a reason to re-score this one.

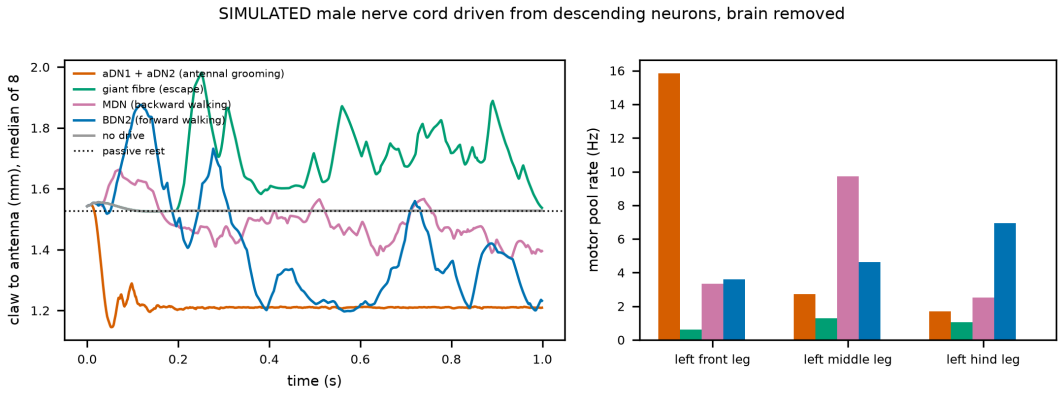


Figure 1. *The SIMULATED male nerve cord, brain removed, driven from descending neurons: claw-to-antenna distance of the FlyMimic leg, and the motor pools of the three left legs, per drive.*

Stage 1d: all four registered predictions pass, on fresh seeds. Read as the time the claw spends near the head and as the rhythmicity of its movement, the two drives separate cleanly: under grooming drive the claw is near the head for 0.9665 of the second, under forward-walking drive for 0.368, and never under the giant fibre or without drive; forward walking moves the leg more rhythmically (0.4599 against 0.2374); the result holds at every activation scale, and stage 1c's front-leg specificity replicates (front-leg pool 16.066 Hz under grooming drive). Rhythmicity alone does not mark walking: the giant fibre scores as high (0.4561), while keeping the claw away from the head. **Stage 1's question is answered for the nerve cord:** the descending neurons that start antennal grooming in the fly move this leg to the head and hold it there, specifically. Next is the brain: the FlyWire brain joined to this nerve cord through type-matched descending neurons, so that sensory drive, not a stimulated command neuron, starts the movement.

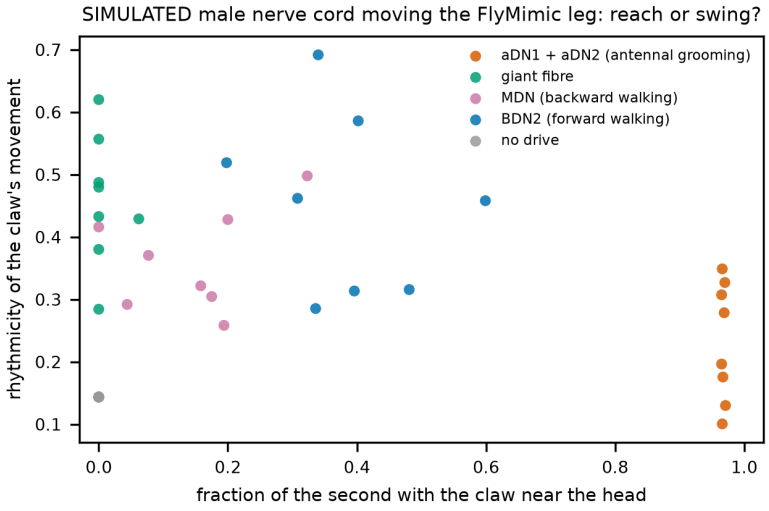


Figure 2. *Time near the head against rhythmicity, one point per SIMULATED trial: the antennal-grooming drive holds the leg at the head; walking and escape drives move it rhythmically and away.*

Stages 1e and 1f: the FlyWire brain drives the male nerve cord, end to end. Each FlyWire descending neuron was paired with the male-CNS descending neuron of the same cell type and side (1,220 of 1,303 matched), the published brain model was run under antennal, sugar or no input, and its descending spikes were replayed into the male nerve cord, which moved the leg. Stage 1e's registered precondition, an absolute 10 Hz line for the grooming descending neurons, failed at 8.531 Hz, so its leg results are uninterpretable; it also showed that its time-near-head readout counted a mild flexion under sugar input as a reach. Stage 1f, registered with a relative precondition and the depth of reach as readout and run on fresh seeds, passes all five predictions. Antennal input brings the claw to 0.5369 mm of the antenna, sugar input to 0.9377 mm, no input leaves it at rest; the front-leg pool carries the movement; and removing the spikes of the two grooming descending neuron types alone (aDN1, aDN2; four neurons) returns the claw to 1.4793 mm. **Stage 1 is complete:** sensory input to a validated whole-brain model, carried by the descending neurons the fly uses for antennal grooming, moves a muscle-driven leg to the head; sugar input does not, and without those four neurons the movement is gone. Next is stage 2 (feeding and walking) and stage 3 (closing the loop).

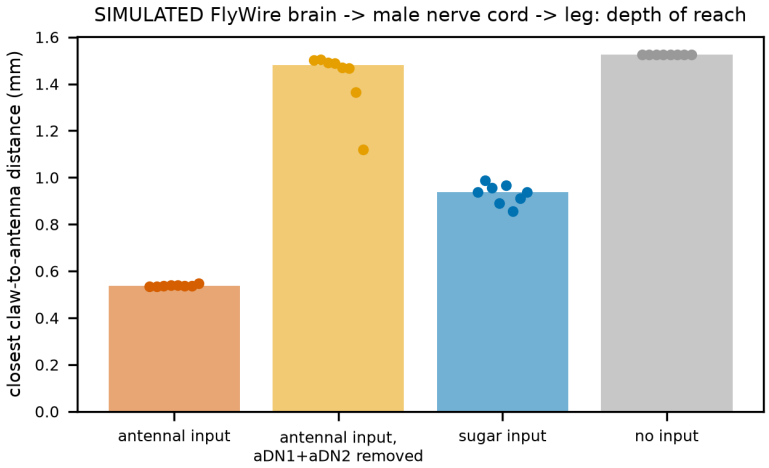


Figure 3. *Depth of reach per condition in the SIMULATED relay (FlyWire brain -> matched descending neurons -> male nerve cord -> FlyMimic leg), one point per trial: antennal input reaches deepest; removing aDN1 and aDN2 abolishes the reach; sugar input flexes the leg only partway.*

Stage 2a: no walking rhythm from the nerve cord alone (failed as registered). Before any body, stage 2a asked whether the male nerve cord turns its forward-walking command (BDN2, DNg100) into a motor rhythm: tibia flexors and extensors alternating within a leg, the two front legs alternating, and a tripod across legs, read from the twelve tibia motor pools of the six legs. Without drive the cord is silent (W0 passes). Under the walking command it is active but tonic: the left middle-leg tibia extensors fire at 54.12 Hz and the right hind-leg extensors at 34.12 Hz, while the tibia flexor pools of five of the six legs stay silent (left front flexors 0.0 Hz). The left front flexor pool has no rhythm (median rhythmicity 0.0), so no phase can be measured and W1 to W4 fail. Backward-walking drive (MDN) gives the same picture with different legs. This is the risk §6 names: a leaky integrate-and-fire nerve cord, with connectome weights and no intrinsic currents, has no half-centre oscillator. The next step is the one §6 commits to, a stated model change registered before it runs: add the intrinsic property rhythm generation needs (spike-frequency adaptation) and test again on fresh seeds, with the connectome-only result kept as the reference.

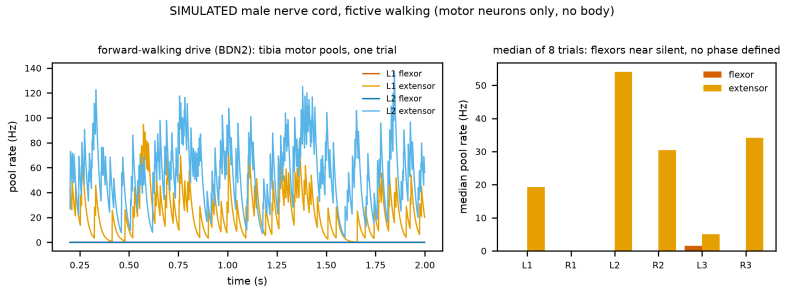


Figure 4. *Tibia flexor and extensor pool rates on the six legs under the walking command, SIMULATED fictive walking in the male nerve cord: extensors fire tonically, flexors stay silent, and there is no alternation.*

Stage 2b: sugar input to the FlyWire brain extends a simulated proboscis; bitter input gates it (six of six registered predictions pass). Every proboscis motor neuron that McKellar et al. (2020) tie to a muscle has a FlyWire cell type (through the male-CNS cross-reference). The rostrum protractor (MN9) and its retractors (MN1, MN2) set the rostrum's command; the haustellum extensors (MN4) and flexors (MN3) set the haustellum's. Both commands drive the proboscis of the flybody whole-fly model. Under sugar input the extending pools reach activations of 0.9644 (rostrum) and 0.8858 (haustellum), their antagonists 0.0561 and 0.0, and the proboscis extends fully (0.3431 mm peak; full extension is 0.3428 mm). Bitter input alone, antennal input and no input leave it at rest. Adding bitter input to sugar input halves the peak (0.1618 mm) and cuts the time-averaged extension from 0.3054 to 0.0488 mm: the proboscis starts to extend, then is held back to brief flicks. The result holds at the other activation scales. Only the MN9 part of this was seen before registration (it is the published model's result); the haustellum pools and their antagonists were not. Exploratory, not scored: sugar input recruits only feeding motor neurons (neck, eye, antennal and crop motor neurons stay silent), and the bitter gate acts on the proboscis pools (other proboscis motor neurons 52.25 Hz under sugar, 2.75 with bitter added) far more than on the ingestion pool (24.58 and 19.98 Hz). **Stage 2's feeding test passes; its walking test (2a) does not.** The body is held fixed and its joints are position servos, so this is the motor command made visible, not a muscle-level proboscis.

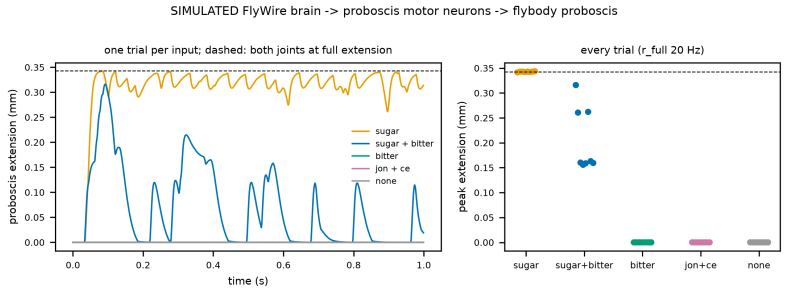


Figure 5. *Proboscis extension in the SIMULATED FlyWire brain -> proboscis motor neurons -> flybody chain: sugar extends the proboscis fully and holds it; adding bitter input lets it start and then holds it back; bitter, antennal or no input leave it at rest.*

Stage 2c: adding spike-frequency adaptation does not make the nerve cord walk (stopped as registered). §6 committed to adding an intrinsic property if stage 2a failed. Adaptation's two parameters have no published fly value, so they were swept over a registered grid of 36 cells. The registration fixed in advance a rule for choosing one cell and a confirmatory test on fresh seeds, with a rewired control and a comparison with the published step period. No cell passes any of the four walking tests (rhythm: 0 cells), so the rule selects nothing, the stage stops, and its seven predictions fail. Adaptation does act: the tibia extensor pools fall from 21.3 Hz (weakest, fastest adaptation) to 0.55 Hz (strongest, slowest). But the flexor pools never exceed 0.22 Hz in any cell, so there is nothing for the extensors to alternate with. An exploratory diagnostic (not registered; exploration seeds) shows why. With every inhibitory connection removed the flexor pools fire at 151.341 Hz, so excitation does reach them. Of the neurons the walking command activates, each flexor motor neuron receives 274.73 inhibitory against 130.78 excitatory synapses, while each extensor motor neuron receives 754.17 excitatory against 698.17 inhibitory. GABA alone (182.68 per flexor neuron) outweighs that excitation. Whether the flexors stay net-inhibited if glutamate excites depends on the model's rule that every glutamate synapse inhibits (92.05 per flexor neuron). Under the model's signs, the walking command's population holds the flexors down and drives the extensors. A rhythm then needs something that periodically releases the flexors, which uniform adaptation does not provide. Candidates for a registered next step are the glutamate sign rule (glutamate excites through some receptors in the fly nerve cord), sensory feedback from the leg (stage 3), or a different intrinsic property (post-inhibitory rebound).

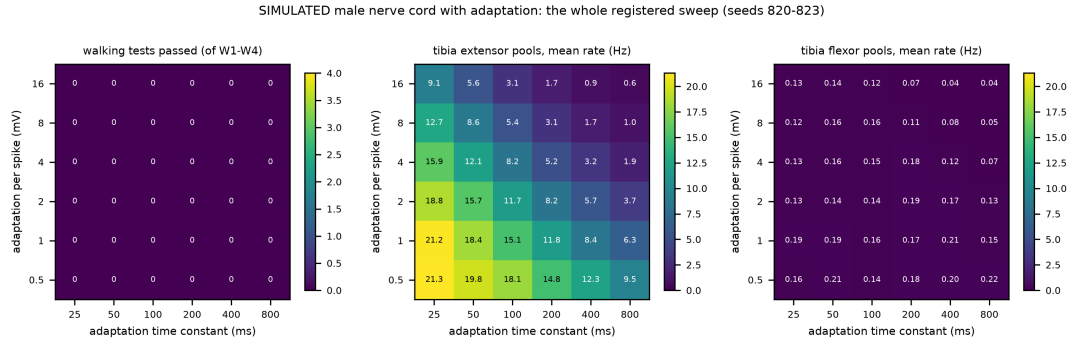


Figure 6. *The whole registered SIMULATED adaptation sweep: no cell passes a walking test; adaptation lowers the tibia extensor pools smoothly, while the flexor pools stay near zero everywhere.*

8. What carries over unchanged

Pre-registration before every run; licensed data with checksums in the manifest; every reported number written by a script and checked against the documents that cite it; failed predictions reported beside the passed ones; and a demo anyone can open.